

Thesis Defense Study Cards

Data-Driven Cognitive Phenotyping in Acquired Brain Injury
20-minute defense rehearsal cards

How To Use These Cards

Use each card as a prompt, not as a script. Aim to answer each “Front” from memory in 20-40 seconds, then check the “Back”. For the examiner cards, practice giving a calm two-part answer: first the direct answer, then the methodological reason.

Core Message Cards

Card 1. One-Sentence Thesis

Front: What is the thesis in one sentence?

Back: In incomplete routine neuropsychological datasets after acquired brain injury, unsupervised learning identifies a robust cognitive severity gradient rather than a hidden structure of sharply separated cognitive subtypes.

Card 2. Main Contribution

Front: What is the main contribution of the dissertation?

Back: I combine missingness-aware imputation, domain-level aggregation, UMAP, HDBSCAN, and sensitivity testing to show that routine clinical neuropsychological data can support clinically interpretable severity stratification.

Card 3. Conservative Interpretation

Front: What is the safest interpretation of the clusters?

Back: They are ordered severity tiers, not natural kinds. The tiers are useful bands along a cognitive severity continuum.

Card 4. Why Missingness Matters

Front: Why is missingness central to the thesis?

Back: Missingness is clinically structured. It can reflect fatigue, impairment, test tolerance, clinician judgement, or practical assessment limits. Removing incomplete rows would discard many clinically important assessments.

Slide Cards

Card 5. Slide 1 - Opening

Front: What should Slide 1 establish?

Back: The dissertation asks whether routine neuropsychological data can reveal reliable cognitive organization after ABI. The answer is yes, but the main structure is a severity gradient, not sharp subtype separation.

Key phrase: “The strongest signal is cognitive severity.”

Card 6. Slide 2 - Thesis In One Sentence

Front: What are the three ideas inside the thesis sentence?

Back: Diagnosis does not fully explain cognitive variability. Missingness must be handled because it is clinically meaningful. The recovered tiers remain robust across imputation strategies.

Card 7. Slide 3 - Clinical Problem

Front: Why is ABI difficult to classify cognitively?

Back: ABI affects multiple domains, including attention, memory, language, executive function, visuospatial perception, and orientation. Patients with the same diagnosis can have different rehabilitation needs.

Card 8. Slide 3 - Research Question

Front: What is the research question?

Back: Can data-driven clustering produce clinically meaningful cognitive organization, and can that organization be trusted when missing data are managed rather than ignored?

Card 9. Slide 4 - H1

Front: What does H1 test?

Back: H1 tests whether discrete and stable clusters can be recovered. Criteria include silhouette greater than 0.40, less than 30% noise, and at least two clusters for most imputation methods.

Card 10. Slide 4 - H2

Front: What does H2 test?

Back: H2 tests robustness to imputation using mean pairwise adjusted Rand index greater than 0.50. It uses a fixed-reference design, so it tests whether imputed points land similarly under a fixed embedding and partition.

Card 11. Slide 4 - H3

Front: What does H3 test?

Back: H3 tests whether the cognitive severity tiers are clinically relevant by examining their association with clinical unit assignment, using chi-square statistics, Cramer's V, and diagnosis-adjusted analysis.

Card 12. Slide 4 - H4

Front: What does H4 test?

Back: H4 tests whether domain-level aggregation improves the analysis compared with using all individual neuropsychological variables.

Card 13. Slide 5 - Cohort Numbers

Front: What are the key cohort numbers?

Back: 17,406 assessments, 7,285 patients, fourteen neuropsychological variables, six cognitive domains, and 53.1% complete rows.

Card 14. Slide 5 - Why Complete Cases Are Not Enough

Front: Why not use only complete cases?

Back: Complete-case analysis would discard about 46.9% of assessments. That is not just loss of power; it can change which patients are represented.

Card 15. Slide 6 - Missingness Mechanism

Front: How should missingness be described?

Back: Missingness reflects the clinical testing process rather than random administrative failure. It can indicate fatigue, disorientation, motor limitations, low tolerance, or clinician judgement.

Card 16. Slide 7 - Pipeline

Front: What is the analysis pipeline?

Back: Impute missing data, scale scores, aggregate into six domains, embed with UMAP, cluster with HDBSCAN, and evaluate separation and noise.

Card 17. Slide 7 - H2 Fixed Reference

Front: What is the fixed-reference design?

Back: MICE defines the anchor scaler, UMAP embedding, and HDBSCAN cluster configuration. Other imputation methods are projected through that fixed structure to isolate the effect of imputed values.

Card 18. Slide 8 - H1 Main Result

Front: What are the three recovered tiers?

Back: Above-Average, Near-Normal, and Global Impairment.

Card 19. Slide 8 - Tier Sizes

Front: What are the tier sizes?

Back: Above-Average: 6,725 assessments. Near-Normal: 6,328 assessments. Global Impairment: 4,353 assessments.

Card 20. Slide 8 - Interpretation Of Tiers

Front: Are the tiers discrete phenotypes?

Back: I do not interpret them as sharply separated phenotypes. I interpret them as ordered severity bands along a continuous cognitive axis.

Card 21. Slide 9 - PCA Evidence

Front: What does the PCA show?

Back: The first principal component explains nearly 56% of variance, and all domains load positively. This supports a broad cognitive severity dimension.

Card 22. Slide 9 - Severity Profile

Front: What is the strongest signal in the data?

Back: Global cognitive severity. The most prominent signal is not a symptom-specific dissociation or a catalogue of distinct subtypes.

Card 23. Slide 10 - H2 Result

Front: What is the main H2 result?

Back: The mean pairwise ARI is .71, with a 95% bootstrap confidence interval of .70 to .72. This indicates substantial agreement across imputation methods under the fixed-reference design.

Card 24. Slide 10 - H2 Caveat

Front: What is the important limitation of H2?

Back: H2 supports robustness of point assignments under a fixed embedding and partition. It does not prove that every independently refitted UMAP plus HDBSCAN pipeline recovers the same density hierarchy.

Card 25. Slide 11 - Complete-Case Result

Front: What happens with complete-case analysis?

Back: Only 53.1% of assessments are retained. The complete-case baseline recovers two clusters rather than the desired three-tier solution, and its silhouette is lower, around .26 versus about .53 for the MICE reference.

Card 26. Slide 12 - H3 Result

Front: What is the clinical association result?

Back: Tier membership is associated with clinical unit assignment. The chi-square statistic is 911, the p-value is on the order of 10^{-156} , and Cramer's V is 0.16.

Card 27. Slide 12 - Diagnosis Adjustment

Front: Does diagnosis explain the clinical-unit association?

Back: Not fully. The Cochran-Mantel-Haenszel test shows the association persists after controlling for diagnosis, so severity tier adds information beyond diagnosis.

Card 28. Slide 13 - H4 Result

Front: Why use domain-level scores?

Back: Domain-level scores improve numerical stability and interpretability. Mean VIF decreases from about 2.80 to about 1.90, and the condition number improves from about 62 to about 12.

Card 29. Slide 13 - Silhouette Difference

Front: How different is the silhouette for domains versus individual variables?

Back: The difference is small but favorable to domains: about .48 for domains versus .47 for individual variables.

Card 30. Slide 14 - Four Hypotheses Summary

Front: How do the four hypotheses end?

Back: H1 is supported as severity tiers, H2 is supported within the fixed-reference design, H3 is supported through clinical-unit association, and H4 is supported through improved numerical properties of domain-level clustering.

Card 31. Slide 15 - Interpretation

Front: What is the central interpretation?

Back: The data support a continuum of cognitive severity rather than a set of discrete cognitive subtypes.

Card 32. Slide 15 - Subtype Caution

Front: Does the study prove that domain-specific dissociations do not exist?

Back: No. It shows that in this dataset and pipeline, the strongest recoverable signal is overall cognitive severity. Domain-specific dissociations may still exist in other designs or subgroups.

Card 33. Slide 16 - Clinical Utility

Front: How can clinicians use the tiers?

Back: The tiers can support discussion of rehabilitation intensity and can be reported with confidence scores. They should supplement clinical judgement, not replace it.

Card 34. Slide 16 - Analytic Utility

Front: How can researchers use the pipeline?

Back: Researchers can adapt it as a missingness-aware framework for studying relationships between cognitive severity and outcomes after acquired brain injury.

Card 35. Slide 17 - CogDash

Front: What is CogDash?

Back: CogDash is a Streamlit dashboard that makes the analysis explorable through a cluster explorer, patient lookup, new-patient classifier, hypothesis summary, and advanced analytics tools.

Card 36. Slide 17 - CogDash Boundary

Front: What should CogDash not be framed as?

Back: It should not be framed as an automated clinical decision-maker. It is a transparent decision-support and exploration tool.

Card 37. Slide 18 - Limitations

Front: What are the main limitations?

Back: Single-site data, no full longitudinal recovery model, MAR assumptions for imputation, possible MNAR mechanisms, and no functional outcomes such as return to work or length of stay.

Card 38. Slide 19 - Conclusion

Front: What are the three final conclusions?

Back: Routine ABI neuropsychological data contain a cognitive severity gradient. Imputation is a substantive modelling decision. Domain-level severity tiers are clinically interpretable and numerically stable.

Card 39. Slide 20 - Closing

Front: What is the final message before questions?

Back: The analysis identifies something clinically useful but conservatively interpreted: an imputation-robust cognitive severity continuum after acquired brain injury rather than discrete cognitive subtypes.

Examiner Question Cards

Card 40. Why HDBSCAN?

Front: Why did you use HDBSCAN rather than k-means?

Back: HDBSCAN can identify dense regions and label noise without forcing every observation into a centroid. That matters in clinical data where borderline or atypical cases may not belong cleanly to a cluster.

Card 41. Why UMAP?

Front: Why use UMAP before clustering?

Back: UMAP provides a nonlinear embedding that preserves local neighborhood structure while reducing dimensionality. It helps expose structure in domain-level cognitive data, but I treat UMAP-based silhouette cautiously because the embedding can exaggerate separation.

Card 42. Is The Silhouette Optimistic?

Front: Is the reported silhouette possibly optimistic?

Back: Yes. Silhouette in UMAP space can be upward biased because UMAP is designed to emphasize neighborhood separation. I therefore interpret silhouette as an internal pipeline-quality statistic, not proof of natural categories.

Card 43. Does H2 Prove Full Imputation Invariance?

Front: Does H2 prove the discovered structure is completely imputation-invariant?

Back: No. H2 tests whether point assignments are robust under a fixed embedding and partition. It does not claim that independently rerunning the full pipeline always recovers identical density hierarchies.

Card 44. Why Not Complete-Case Analysis?

Front: Why is complete-case analysis not acceptable here?

Back: It removes nearly half of the analyzed assessments and may preferentially remove more impaired patients. In this study it also changes the recovered clustering structure.

Card 45. Why Domain Aggregation?

Front: What is the argument for domain aggregation?

Back: Aggregation reduces redundancy, improves numerical conditioning, and creates clinically interpretable variables. It sacrifices some test-level detail, but the gain in stability and interpretability is valuable for this analysis.

Card 46. What If Real Subtypes Exist?

Front: Could real cognitive subtypes still exist?

Back: Yes. The thesis does not rule them out. It shows that this dataset and pipeline primarily recover a severity gradient. More targeted designs may detect domain-specific dissociations.

Card 47. What Does Clinical Relevance Mean Here?

Front: What does clinical relevance mean in H3?

Back: It means tier membership is statistically associated with clinical unit assignment and remains associated after diagnosis adjustment. It does not mean the model determines clinical placement.

Card 48. What Is The Practical Output?

Front: What would a practical user get from this work?

Back: A severity tier, a confidence estimate, and interpretable domain-level information that can support rehabilitation discussion and research analysis.

Card 49. What Is The Strongest Defense Of The Thesis?

Front: What is the strongest defense of the thesis?

Back: Multiple lines of evidence converge: stable tier recovery, PCA showing a broad severity dimension, imputation robustness under fixed reference, clinical-unit association, and better numerical properties from domain aggregation.

Card 50. What Is The Most Important Caveat?

Front: What caveat should you be ready to say clearly?

Back: The clusters should not be overclaimed as discrete phenotypes. The strongest justified conclusion is severity stratification along a continuum.

Rapid Number Drill

Prompt	Answer
Assessments analyzed	17,406
Patients	7,285
Neuropsychological variables	14
Cognitive domains	6
Complete rows	53.1%
Incomplete rows	46.9%
Above-Average tier	6,725
Near-Normal tier	6,328
Global Impairment tier	4,353
PC1 variance	nearly 56%
H2 mean ARI	.71
H2 bootstrap CI	.70 to .72
High-confidence consensus labels	over 86%
Complete-case silhouette	about .26
MICE reference silhouette	about .53
H3 chi-square	911
H3 Cramer's V	0.16
H4 domain silhouette	.48
H4 variable silhouette	.47
Mean VIF change	about 2.80 to 1.90
Condition number change	about 62 to 12

Final Rehearsal Mantra

The analysis does not claim to discover sharp cognitive subtypes. It supports a robust, clinically interpretable severity gradient that remains meaningful when missingness is handled explicitly.